BIOT 6900 Deliverable 1
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1. Introduction

Alzheimer's Disease (AD) is a prevalent condition that affects approximately 1 in 9 people 65 and older in the United States [0]. It is the most common form of dementia, and is ultimately fatal. It can cause memory loss, language problems, personality changes, and agitation. I personally have lost a family member to AD, and seeing the effect it has on the person suffering and their loved ones, is heartbreaking.

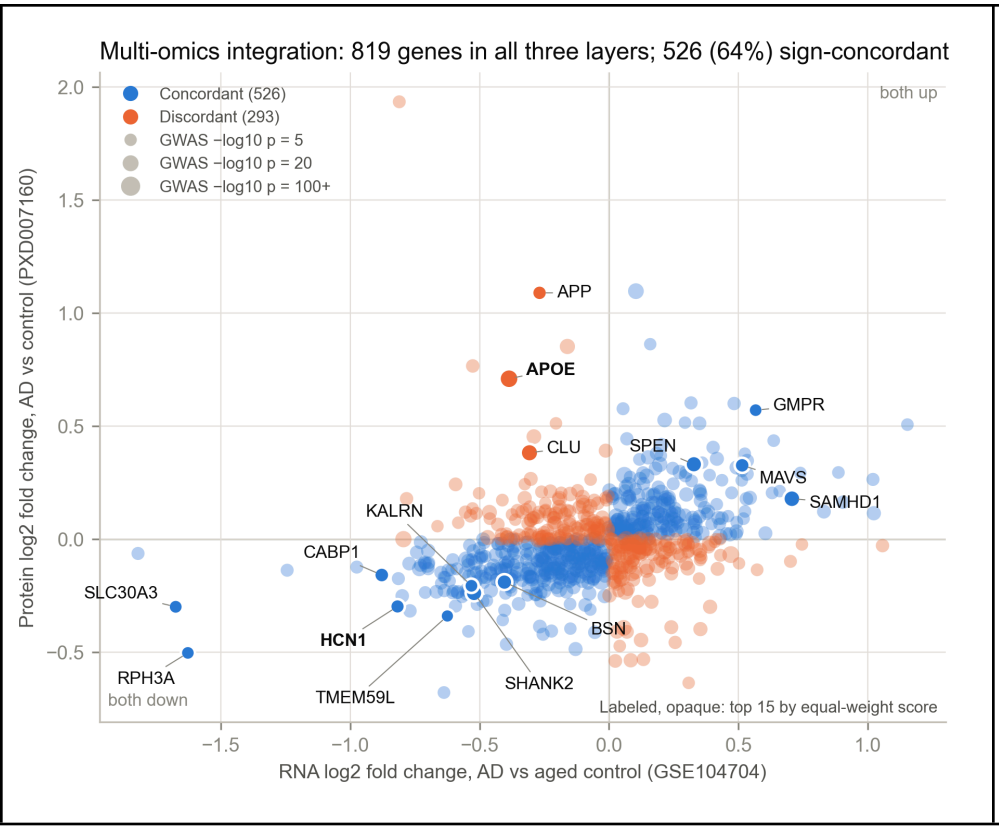
Given this, research into this disease is crucial to many, myself included. In this paper, I seek to find a way to medicate against AD, by asking the question: which genes have multi-omic evidence to link them to AD, sequence based tractability, and structural tractability, supporting them as a target against Alzheimer's?

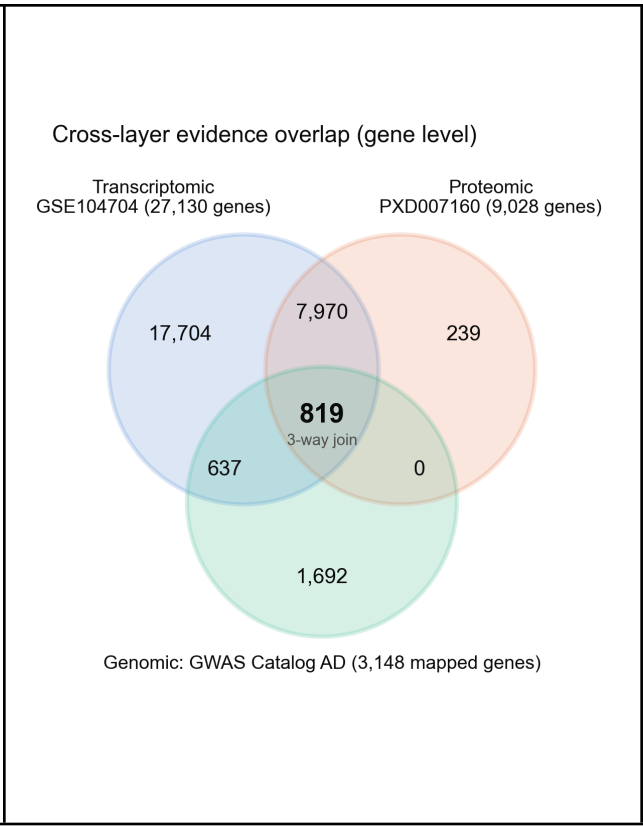
To begin, I had to define what makes a good target. A good target is relevant to the disease, and has sequential and structural tractability. First, I looked at multi-omics data: genomic, transcriptomic, and proteomic, to determine which genes are relevant to AD. Then, I looked at sequence-based tractability by including a prior and ranking them on tractability scores. Finally, I tested the sequence-based evidence with structural evidence in order to conclude which genes are the best possible candidates. As shown below, hyperpolarization-activated cyclic nucleotide-gated channel 1 (HCN1) was a strong potential target throughout the analyses, whereas apolipoprotein E (APOE) was a strong multi-omic contender that was less strongly evidenced by sequence and structure.

2. Multi-omics

For the multi-omics data, I used three sources, one for each data type. For transcriptomics, I used GSE104704, which studied the temporal lobe for 12 AD patients and 10 controls [1]. For proteomics, I used PXD007160, which studied the frontal cortex for AD vs control [2]. Finally, for genomics, I used the GWAS catalog entry MONDO_0004975 [3]. I had to integrate at the gene level, given the data was from 3 different sources. I integrated at equal weights, since genomics gives the important information about underlying causes for development of AD, but the context given by transcriptomics and proteomics of how the expression actually affects potential AD is ultimately more valuable, meaning ⅔ is a fair portion to assign to this context.

After joining the datasets, there were 819 genes left. The breakdown of how many were in each section is shown in Figure 2 below. Then RNA log2 change, protein log2 change, and a negative log10 p-value for GWAS were found and used to create a ranking of the genes. Figure 1 shows these values and highlights the top 15 ranked genes when the equal weighting between these scores is applied.





Figures 1&2 1. Multi-omics rankings 2. Overlap of genes in the three datasets

Multiple of the most well known genes related to AD were in the top 15: APOE[4] was ranked #1, CLU[6] #10, and APP[5] #13. Some well-known AD risk markers were ranked lower: MAPT[4] was ranked #45, and BIN1, SORL1 and PICALM[7] were outside the top 50. TREM2 was lost entirely because the proteomics data didn't include it.

3 of the top 15 ranked genes were discordant, and 36% of all the tested genes were discordant as well. Among them was the top-ranked APOE with RNA score -0.39 and protein score 0.71. This discordance in high-ranking genes could be a result of taking the absolute value of the scores, or it could be that different biological pathways lead to different types of effects from the same gene. For example, APOE is involved in several mechanisms that correlate with AD. The lipidation levels of APOE can cause issues with both the clearance of Aß and the pathology of tau, both factors in AD[8]. Given these biological pathways can be complex and varied, it makes sense that discordance could arise in genes that are related to AD.

One non-obvious hit was HCN1, which was ranked second. It is an ion channel that allows sodium and potassium ions through in the process of chemical signalling in neurons[10][13]. Given this, it is plausible that it is related with AD, as is suggested given the ranking.

3. Sequence-based Tractability

This portion was done with ESM-C embeddings using a logistic regression prior[12]. A weight of 0.65 was used in favor of the multi-omics score because the prior was found to be only moderately accurate, with a family-split AUROC of 0.73. Given this, the disease-specific rankings were given a higher weight. The average precision for the family split was 0.548.

Gene	p_sm	p_ab	Final rank
HCN1	0.879	0.005	1
SAMHD1	0.830	0.003	4
SLC30A3	0.414	0.003	3
SHANK2	0.337	0.000	2

Table 1

HCN1's ranking went up to first, with a score of 0.957. It has a resolved structure [13] and an ivabradine pocket that is confirmed by experiment [14]. APOE dropped from first to sixth, given a low p_sm of 0.002. CLU had the highest p_ab of 0.128.

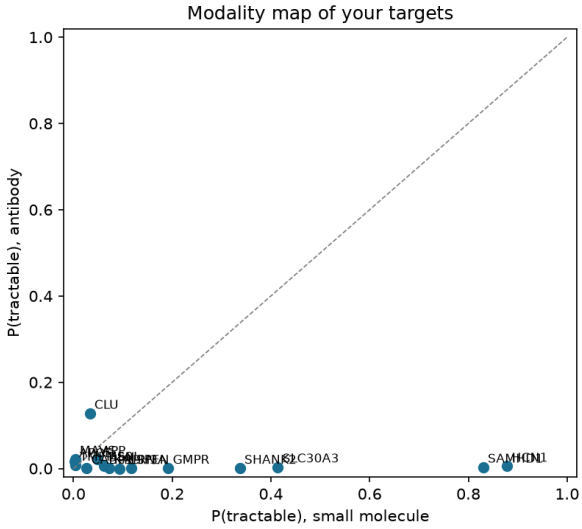


Figure 3 Tractability for antibody vs small molecule modes

4. Structural Evidence

For the structural analysis, the AlphaFold models were used [16], pockets were found with fpocket[17], and the models were compared with experimental structures, 8Y60 for HCN1[14] and 1GS9 for APOE[18].

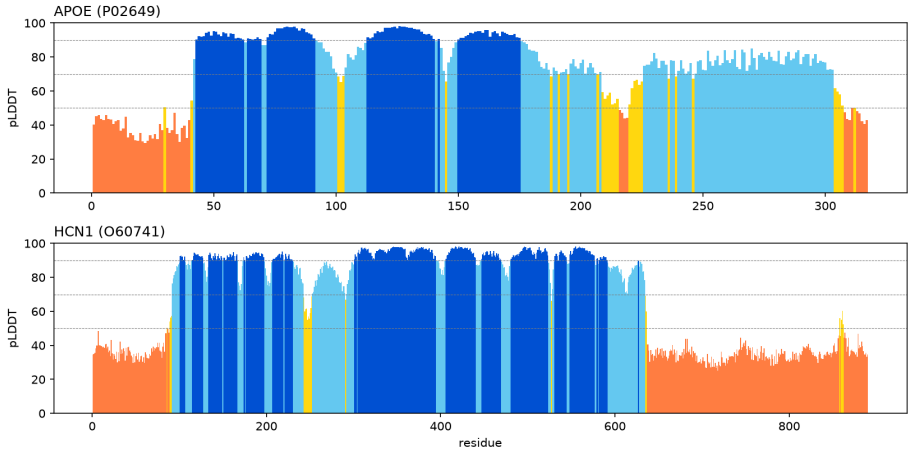


Figure 4 pLDDT for APOE and HCN1

The pLDDT is AlphaFold's measure of confidence in residues, and the values are shown in Figure 4. For HCN1, the mean pLDDT was 68.9, a number that is heavily lowered by the disordered tails at either end of the protein. The confident domain was between residues 92-634, which includes the HCN domain, S1-S6, the C-linker, and the CNBD [10][13]. Over the 459 residue pairs that could be compared with 8Y60, the RMSD was 1.35 Å, which means the residues closely matched the experimental study. Of the pockets found, pocket 2 scores the best at 0.878. However, ivabradine is found in pocket 7, which is ranked second, with a score of 0.418 and 75% coverage [14].

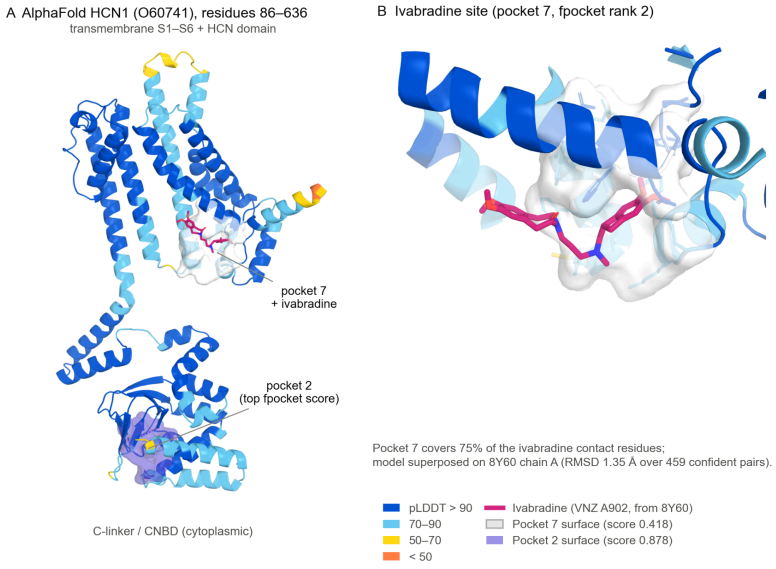


Figure 5 A. AlphaFold structure of HCN1 residues, colored by pLDDT B. ivabradine from 8Y60 placed on pocket 7

APOE had two confident domains, 42-208 and 226-303. Between the domains, the PAE was 25.8 Å. The RMSD was 0.77 Å over 134 pairs, after an 18-residue offset was applied, because the protein is cleaved upon secretion, so 18 residues are removed [19]. The best-ranked pocket, pocket 20, includes low-confidence residues (31-39), and 1GS9 has no ligand [18]. This shows that despite the low confidence, since the pocket uses an averaged pLDDT, the disordered residues are hidden. In the end, this APOE pocket was called to be a possible candidate, but uncertain.

gene	pocket	Druggability Score	Volume	mean_pLDDT	PAE_within	verdict	rank
APOE	20	0.546	910.878	76.5445	4.0125	keep	1
APOE	5	0.005	382.336	73.8571	4.8163	keep	3
APOE	8	0.005	618.009	56.3483	7.5903	reject	4
APOE	2	0.005	250.06	88.07143	1.8775	keep	2
APOE	3	0.002	236.569	71.9000	4.0800	keep	5
HCN1	2	0.878	407.456	92.6588	2.0796	keep	1
HCN1	7	0.418	487.998	84.7453	3.2267	keep	2
HCN1	1	0.245	1159.976	86.3100	3.6825	keep	3
HCN1	8	0.212	544.447	92.7456	1.9844	keep	4
HCN1	6	0.128	384.73	83.4373	4.0222	keep	5

Table 2

5. Prioritized Target Package

The top candidate is HCN1, given that it was ranked highly in multi-omics score, had sequence-based tractability, and had structural tractability of a pocket in pocket 7. Pocket 7 is experimentally shown to have ivabradine, so there is evidence of a small molecule ligand in that region for druggability. One major drawback is that ivabradine is an inhibitor of HCN1, and HCN1 is down in AD. It isn't clear if inhibiting HCN1 is a cause or effect of AD, so further research is needed, but HCN1 is the strongest tractable candidate in this study.

The second target I chose to continue investigating was APOE. It ranked first in multi-omics score, indicating it is strongly related to AD. It ranked lower in sequence-based tractability, however, and its best pocket contained a low-confidence region. The evidence from sequence-based and structure-based tractability was not very convincing, but it remains a possible candidate.

Tier	Gene	M2 rank (score)	RNA / protein log2FC	Concordant	GWAS −log10 p	p_sm / p_ab	M3 rank	Structure	Call
1	HCN1	2 (0.922)	−0.82 / −0.30	Yes	12.4	0.879 / 0.005	1	pLDDT 68.9; drug site = pocket 7 (0.418, 75% of ivabradine site)	Advance
2	APOE	1 (0.924)	−0.39 / +0.71	No	496.4	0.002 / 0.016	6	pLDDT 75.5; pocket 20 (0.546), partly low-confidence	Genetic anchor; non-small-molecule
3	SAMHD1	7 (0.902)	+0.71 / +0.18	Yes	35.7	0.830 / 0.003	4	Not assessed	Next for AlphaFold
Watch	SLC30A3	5 (0.911)	−1.67 / −0.30	Yes	10.7	0.414 / 0.003	3	Not assessed	Lower priority
Watch	SHANK2	3 (0.914)	−0.52 / −0.24	Yes	30.0	0.337 / 0.000	2	Not assessed	Lower priority

Table 3

6. Limitations and validation

There were some limitations with the findings in this paper. Firstly, the original multi-omics data came from different cohorts of people, and from different portions of the brain [1][2]. This limits claims that can be made to correlation, and nothing can be said about each layer occurring together, since they were measured in different people. Another limitation is that sequence prior does not score pathogenicity. ε4 (C130R) scores highly positive but is known to raise risk, while ε2 (R176C) scores very negative but is known to be protective [15]. Another limitation is the fact that the membrane is absent from the structure model of HCN1 [14] and the lipid is missing from the structure model of APOE [19], both of which are external factors which highly affect the functionality of the proteins. Finally, only two targets have been examined for structural evidence so far, so another potential candidate could end up being promising.

To validate the HCN1 results, an experiment that included the membrane would be needed, as well as an experiment to test if a reduction in the amount of HCN1 would improve or worsen AD.

In conclusion, HCN1 was the primary target, APOE was the known risk factor to potentially consider, and SAMHD1 is the next-ranked potential candidate for structural analysis.

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